

PHOTON ABSORPTION AND ENTANGLEMENT SEVERANCE

as a Unified Origin of Dark Matter and Dark Energy

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Nine-paper corpus: The Quantum Soul (DOI: 10.5281/zenodo.19911973) | Resonance Substrate Theory (MRME) | Resonance Cosmology | The Periodic Table of Universes | Multiverse Mathematics | This paper | Stress as Resonant Catalyst | The Reflective Medium | Time as Resonant Interval

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Epistemic Level Guide

Label	Meaning
[M]	Mathematical — internally consistent within general relativity and quantum mechanics; not speculative.
[P]	Physical — a speculative claim that $T^{\text{uv}}_{\text{dark}}$ is real and corresponds to observable dark phenomena; requires observational confirmation.
[Mech]	Mechanistic — highly speculative; extends beyond standard QED and requires theoretical derivation from a fundamental Lagrangian.
[RC]	Corpus connection — an explicit link to a companion paper in the nine-paper corpus; shows where this mechanism sits in the broader framework.

ABSTRACT

Photon absorption is not annihilation but severance — the origin of the dark sector. Dark matter and dark energy together constitute approximately 95% of the universe's total energy budget, yet neither has a confirmed mechanistic origin. This paper proposes a unified origin based on photon absorption and entanglement severance, situated explicitly within the nine-paper Resonance Cosmology corpus. In the corpus framework, the ether substrate — the mathematical meta-space established in the Mathematical Resonance Model of Ether (MRME) — provides the medium through which the dark field ϕ_{dark} operates. The dark field ϕ_{dark} is not an arbitrary new entity; it is identified as an E1-band excitation of the ether substrate, a resonance mode of the quantum vacuum band whose existence is motivated by the MRME's seven-band classification. Physical constants, including the effective cosmological constant Λ_{eff} , are derived quantities in the Resonance Cosmology framework — eigenvalues of the ether operator — not fundamental parameters. This paper provides the specific physical

mechanism by which Λ_{eff} is generated: the integrated history of diffuse photon absorption over cosmic time, accumulating dark energy into the E1-band ether density at rate ϵ per absorption event. This resolves the 123-order-of-magnitude cosmological constant discrepancy not by cancellation but by reframing: the QFT vacuum energy prediction and the observed cosmological constant refer to different physical quantities within different bands of the ether substrate. The severance mechanism itself rests on three pillars. First, photon emission produces a momentum-entangled joint state between the photon and its source, following from conservation of momentum in quantum mechanics. Second, upon absorption, the gravitational field and mass of the absorbing object sever this quantum correlation, releasing a latent dark contribution not captured by standard stress-energy accounting, governed by the severance operator $S = \epsilon(1 - \sqrt{g_{00}})$. Third, the outcome is density-dependent: in dense media the severed state localises into pressureless dust ($w = 0$, dark matter); in diffuse media it couples to vacuum field modes and produces negative-pressure contribution ($w = -1$, dark energy). A consolidated photon budget across four cosmic populations yields $\epsilon \sim 10^{-28}$, consistent with producing both $\Omega_{\text{DM}} \sim 0.27$ and $\Omega_{\text{DE}} \sim 0.68$. Six falsifiable predictions are presented. A fourth epistemic label [RC] is introduced throughout to mark explicit corpus connections.

Six Falsifiable Predictions

- 1. Spatial correlation between photon absorption zones and dark matter halos (Euclid + ALMA/Herschel cross-correlation).
- 2. Galactic core dark matter excess in absorption-rich versus absorption-poor galaxies (SPARC rotation curves).
- 3. CMB angular power spectrum anomalies at $l \sim 1000-3000$ from recombination-era severance (CMB-S4, Simons Observatory).
- 4. Dark energy equation of state tracking IGM photon absorption rate over cosmic time (DESI BAO vs. Ly α forest opacity).
- 5. Bullet Cluster kinematic decoupling consistent with zero electromagnetic interaction over pre-collision history (Chandra, Athena).
- 6. Laboratory calorimetric energy deficit proportional to gravitational potential depth at rate $\epsilon \sim 10^{-28}$ per absorption event.

1. Introduction

Corpus context: where this paper sits.

This paper is the sixth of nine in the Resonance Cosmology corpus. The corpus establishes: (1) The Quantum Soul — the ether substrate and its relationship to consciousness; (2) Resonance Substrate Theory (MRME) — five quantities, seven resonance bands E1-E7; (3) Resonance Cosmology — the disciplinary charter, physical constants as eigenvalues of the ether operator; (4) The Periodic Table of Universes — four resonance classes; (5) Multiverse Mathematics —

categorical foundations; (6) This paper — the photon absorption mechanism generating the dark sector; (7) Stress as Resonant Catalyst — consciousness and perturbation; (8) The Reflective Medium — the Theory of Everything programme; (9) Time as Resonant Interval — the temporal dimension of the ether. The dark sector mechanism developed here is grounded in the ether substrate established in papers 1-5. The dark field ϕ_{dark} is an E1-band ether excitation. The effective cosmological constant Λ_{eff} is a derived quantity in paper 3's eigenvalue framework. The temporal accumulation integral (Eq. 12) is the same structure as the temporal period T_1 in paper 9.

The two most abundant components of the observable universe — dark matter and dark energy — share a peculiar epistemic status. Both are inferred from their gravitational effects with high confidence: dark matter from galactic rotation curves ^[1], gravitational lensing ^[2], and the cosmic microwave background power spectrum ^[3]; dark energy from Type Ia supernova luminosity distances ^[4] and baryon acoustic oscillations ^[5]. Together they constitute approximately 95% of the total energy budget of the universe. Yet neither has a confirmed physical origin. The standard cosmological model Λ CDM treats dark matter as pressureless cold dust and dark energy as a cosmological constant Λ , but provides no mechanism for either. Proposed dark matter candidates — WIMPs, axions, and sterile neutrinos — have evaded detection across four decades of dedicated direct searches ^[6,7]. Dark energy remains entirely unexplained at the fundamental level, and the cosmological constant problem ^[16] presents one of the deepest unresolved puzzles in theoretical physics.

Within the Resonance Cosmology framework, this situation is expected. Physical constants — including the effective cosmological constant Λ_{eff} — are not fundamental parameters but derived quantities: eigenvalues of the ether resonance operator, generated by the structure of the ether substrate at its E7 resonance band ^[RC:3]. The cosmological constant problem — the 123-order-of-magnitude discrepancy between the QFT vacuum energy prediction and the observed value — is not a fine-tuning problem within this framework; it is a category error. The QFT prediction refers to the zero-point energy of quantum fields within the E1 band of the ether substrate. The observed Λ_{eff} is the accumulated contribution of diffuse photon absorption to the E1-band ether density over cosmic time. These are different physical quantities, operating in different regions of the ether spectrum, and their apparent discrepancy dissolves when they are correctly identified. This paper provides the specific mechanism: each photon absorption event in a diffuse medium deposits a fraction ϵ of the photon's energy into the E1-band ether density, and the integral of these deposits over 13.8 Gyr of cosmic photon absorption produces the observed Λ_{eff} .

[RC] *This reframing of the cosmological constant problem connects directly to Resonance Cosmology: Framework and Parameters [3] (constants as eigenvalues, Λ_{eff} as derived) and Time as Resonant Interval [9] (temporal accumulation as the ordering principle of ether evolution).*

This paper proposes a unified origin for both dark matter and dark energy grounded in a reinterpretation of photon absorption. In standard quantum electrodynamics, photon absorption is treated as annihilation: the photon ceases to exist as a quantum excitation and its energy is transferred entirely to the absorbing medium. We challenge this accounting. We propose that photon emission produces a momentum-entangled quantum state between the photon and its source — a consequence that follows directly from conservation of momentum in quantum mechanics — and that absorption severs this entanglement through gravitational interaction with the absorbing mass. The severed correlation constitutes a new physical degree of freedom, a dark state, not included in standard stress-energy accounting. This dark state is identified as an excitation of the dark field ϕ_{dark} , which is itself identified as an E1-band mode of the ether substrate established in the MRME [RC-2]. This identification transforms ϕ_{dark} from an arbitrary postulate into a structured hypothesis within a developed theoretical framework.

The dual-outcome character of the mechanism is the paper's central result: the same physical process — entanglement severance at photon absorption — produces dark matter in dense environments and dark energy in diffuse environments. The distinction is determined by whether the severed energy couples to matter modes (dense media, E3-E4 band range) or vacuum field modes (diffuse media, E1 band). This unifies both invisible components of the universe under a single conservation-based framework without requiring particles beyond the Standard Model.

Three epistemic levels are maintained throughout, plus a fourth. [M] Mathematical: the extended conservation law and its integration into Einstein's field equations are internally consistent. [P] Physical: that $T^{\text{uv}}_{\text{dark}}$ is non-zero and corresponds to observable dark matter and dark energy is speculative and requires confirmation. [Mech] Mechanistic: the specific severance mechanism extends beyond standard QED and requires derivation from a fundamental Lagrangian. [RC] Corpus connection: explicit links to companion papers that ground or extend the mechanism.

2. Physical Context

2.1 Photon Physics: Energy-Momentum Without Rest Mass

Photons are the quanta of the electromagnetic field with zero rest mass: $m = 0$. This follows from the general energy-momentum relation in special relativity:

$$E^2 = (pc)^2 + (mc^2)^2 \quad (\text{Eq. 1})$$

For photons, $m = 0$, yielding $E = pc$. A photon carries energy $E = h\nu$ and momentum $p = h\nu/c$. A key consequence is mass-energy equivalence: a photon's energy E is gravitationally equivalent to effective mass $m_{\text{eff}} = E/c^2 = h\nu/c^2$. When photon energy is localised in space by absorption into a bound structure, that localised energy gravitates exactly like a mass equal to E/c^2 , even though the photon carried no rest mass. This resolves the apparent paradox that massless particles can produce mass-like dark states: the dark state's gravitational effect arises from localised energy, not from rest mass.

In the MRME framework [RC:2], energy localisation corresponds to a transition between resonance bands: a photon propagating at E1-E2 band frequencies is absorbed and its energy deposited into a structure operating at E3-E4 band timescales. The severance mechanism operates precisely at this inter-band transition — the moment when E1-band photon energy crosses into the E3-E4 band matter domain. The dark state is the portion of that energy that remains in the E1 band rather than crossing over.

[RC] *The inter-band energy transition interpretation connects directly to the MRME's seven-band classification [RC:2] and to The Reflective Medium's treatment of E1-E4 band boundaries [RC:8].*

2.2 Standard Treatment of Photon Absorption

In QED, photon absorption is described by the interaction Hamiltonian $H_{\text{int}} = -(e/m_e) \mathbf{A} \cdot \mathbf{p}$. The photon annihilation operator removes one quantum from the electromagnetic field while the electron undergoes a transition. This description is extraordinarily accurate for all known electromagnetic phenomena and is not disputed here. Our proposal is an extension, not a replacement: the standard QED accounting captures the energy-momentum transfer from the free-propagating photon field to the absorber, but does not account for an additional degree of freedom — the photon's entangled correlation with its source — that becomes physically significant at the moment of absorption.

[Mech] *This postulate extends beyond standard QED and requires independent theoretical development.*

2.3 The Dark Sector: Properties and Missing Origin

Dark matter is characterised observationally by four properties: (1) it gravitates like matter; (2) it does not interact electromagnetically; (3) it is cold — non-relativistic at the time of structure formation; (4) it clusters on galactic and cluster scales. The strongest direct evidence for dark matter as a component kinematically distinct from baryonic matter comes from the Bullet Cluster [2], where a galaxy cluster collision separated the hot X-ray gas (decelerated by electromagnetic ram pressure) from the dark matter component (which passed through, evidenced by gravitational lensing maps). Dark energy is characterised by a negative equation of state

parameter $w = -1$ (the cosmological constant), spatial homogeneity, and approximately constant energy density as the universe expands.

The cosmological constant problem ^[16] states a 123-order-of-magnitude discrepancy between the QFT vacuum energy prediction ($\sim 10^{94} \text{ g/cm}^3$) and the observed dark energy density ($\sim 10^{-29} \text{ g/cm}^3$). In the Resonance Cosmology framework, this discrepancy is reframed rather than resolved by cancellation: the QFT prediction is the total zero-point energy of quantum fields operating within the E1 band of the ether substrate ^[RC:2]; the observed Λ_{eff} is the accumulated contribution to E1-band ether density from diffuse photon absorption over cosmic history. The two quantities are not in competition — they are different measures of different physical processes within different regions of the ether spectrum. This reframing is developed quantitatively in Section 8.4.

2.4 The Cosmic Photon Budget

The universe contains an enormous number of photons across a range of frequencies and environments. Table 1 summarises the principal photon populations and their relevance to the severance framework.

Photon Population	Number Density	Absorption Environment	Role in Severance Framework
CMB	411 cm^{-3}	Diffuse IGM; occasional dust absorption	Primary candidate for dark energy; diffuse absorption couples to E1-band vacuum modes ($w = -1$). Λ_{eff} accumulation.
Cosmic UV/optical background	$\sim 10\text{-}100 \text{ cm}^{-3}$	Intergalactic gas, dust, galaxy halos	Mixed contribution; both dense (E3-E4 band) and diffuse (E1 band) environments depending on sightline
Stellar photons (galactic disc)	$\sim 10^3\text{-}10^6 \text{ cm}^{-3}$	Gas, dust, molecular clouds	Primary candidate for dark matter; dense absorption couples to E3-E4 band matter modes ($w = 0$). Dark matter halo production.
X-ray background	$\sim 10^{-3} \text{ cm}^{-3}$	Hot intracluster gas, AGN environments	High-energy; strong gravity near AGN significantly enhances $S(G_N, M, r)$. Contributes to both bands near AGN.

Table 1. Photon populations and their roles. MRME band labels added: dense absorption (E3-E4 band range) produces dark matter; diffuse absorption (E1 band) produces dark energy. The band labels connect the photon budget to the ether substrate framework.

[P] Whether the total photon absorption rate integrated over cosmic time produces the observed $\Omega_{DM} \sim 0.27$ and $\Omega_{DE} \sim 0.68$ is addressed quantitatively in Section 6.6.

3. The Source Entanglement Hypothesis

3.1 Quantum Correlations in Standard Emission

In standard QED, the emission of a photon by an atom in excited state $|e\rangle$ produces a transition to ground state $|g\rangle$. When the photon has definite momentum $\hbar k$, the joint state is separable:

$$\Psi = |g\rangle_{\text{source}} \otimes |1_{\mathbf{k}l}\rangle_{\text{photon}} \quad (\text{separable} - \text{standard QED})$$

(State A)

In physical emission processes, however, the photon momentum k is not perfectly definite. By conservation of momentum, if the photon carries momentum $\hbar k$, the source recoils with momentum $-\hbar k$. For a source with initial momentum $|p_0\rangle$, the post-emission state is genuinely entangled:

$$\Psi = \int d\mathbf{k} f(\mathbf{k}) |p_0 - \hbar \mathbf{k}\rangle_{\text{source}} \otimes |\mathbf{k}, l\rangle_{\text{photon}} \quad (\text{Eq. 2})$$

where $f(k)$ is the photon momentum wavepacket amplitude. This correlation is not a hypothesis — it follows directly from the linearity of quantum mechanics and conservation of momentum.

[M] Equation 2 is an established consequence of quantum mechanics. The entanglement arises from momentum conservation at emission and is the same phenomenon exploited in Hong-Ou-Mandel interferometry [20]. The novel hypothesis concerns what happens to this entanglement at absorption.

3.2 The Extended Entanglement Hypothesis

[Mech] The following is a speculative extension beyond standard QED and is explicitly a theoretical postulate requiring future derivation.

We postulate that the momentum-entangled joint state (Eq. 2) is not fully captured by the standard QED stress-energy tensor of the free-propagating photon alone. $T^{\mu\nu}_g$ encodes the free-field energy-momentum of the photon wavepacket; the correlation with the source's recoil state constitutes an additional degree of freedom — the latent correlation — that resides non-locally across the entangled state. When the photon is absorbed by a massive object, we postulate that the gravitational field of the absorber severs this entanglement. The latent correlation, no longer sustained by quantum coherence, must be accounted for in the post-absorption energy

budget. This additional degree of freedom is identified with an excitation of the dark field ϕ_{dark} .

[RC] The dark field ϕ_{dark} is identified as an E1-band excitation of the ether substrate established in the MRME [RC:2]. In the seven-band classification, E1 is the quantum vacuum band — the ground state of the ether at the smallest physical scale. The dark field excitation is a departure from E1-band vacuum order: a localised energy deposit into the ether's most fundamental resonance layer. This identification transforms ϕ_{dark} from an arbitrary new field into a structured component of the existing ether framework.

3.3 Relationship to Decoherence and Gravitational State Reduction

The severance mechanism is structurally analogous to quantum decoherence: a quantum correlation is disrupted by interaction with an environment. Our proposal extends this: we postulate that the photon-source entanglement, when severed by the gravitational/mass interaction, does not simply disperse into conventional thermal degrees of freedom of the absorber. Instead, it produces a new gravitationally active term $T^{\mu\nu}_{\text{dark}}$ — an E1-band ether excitation — not included in standard stress-energy accounting.

This connects naturally to Penrose's proposal ^[21] that quantum superpositions in the presence of significant gravitational curvature are unstable — the gravitational field drives state collapse at a timescale $t \sim \hbar/E_G$. In the MRME framework ^[RC:2], this gravitational decoherence is precisely the mechanism by which a photon transitioning from its propagation timescale (E1-E2 band) to the absorber's matter timescale (E3-E4 band) deposits a residual E1-band excitation. The ether substrate is the medium through which this inter-band energy transfer occurs; the dark state is the E1-band residual of an incomplete inter-band transition.

[RC] The inter-band transition interpretation connects the severance mechanism to The Reflective Medium's analysis of E1-E4 band boundaries [RC:8], where the failure of classical mathematics to bridge E1 and E4 is identified as the structural obstruction to a Theory of Everything. The severance mechanism operates precisely at this boundary.

4. The Severance Mechanism and Conservation Law

4.1 Extended Energy-Momentum Conservation

[M] The tensor equations in this section are mathematically well-defined within general relativity. They extend the physical content of the stress-energy tensor without modifying the geometric structure of the theory.

In general relativity, conservation of energy-momentum is expressed as the covariant divergence of the total stress-energy tensor:

$$\nabla_u T^{uv}_{\text{total}} = 0 \quad (\text{Eq. 3})$$

Our hypothesis adds a third term — the E1-band ether excitation — to the standard two-component decomposition:

$$T^{uv}_{\text{total}} = T^{uv}_g + T^{uv}_{\text{obj}} + T^{uv}_{\text{dark}} \quad (\text{Eq. 4})$$

[P] T^{uv}_{dark} is the physical claim. Its non-vanishing is the central speculative postulate of this paper. Its identification as an E1-band ether excitation [RC:2] is the corpus connection that grounds it.

Given the postulate that T^{uv}_{dark} is non-zero, Eq. 3 requires:

$$\nabla_u T^{uv}_g + \nabla_u T^{uv}_{\text{obj}} + \nabla_u T^{uv}_{\text{dark}} = 0 \quad (\text{Eq. 5})$$

After full absorption ($T^{uv}_g \rightarrow 0$), the remaining balance is:

$$\nabla_u T^{uv}_{\text{obj}} = -\nabla_u T^{uv}_{\text{dark}} \quad (\text{Eq. 6})$$

Equation 6 is a mathematical consequence of Eq. 5 when $T^{uv}_g \rightarrow 0$: whatever energy-momentum is not accounted for in T^{uv}_{obj} (the latent correlation released by severance) appears as the divergence of T^{uv}_{dark} — the E1-band ether density contribution. The magnitude of this residual is governed by the severance operator.

4.2 The Energy Partition Equation

The severance process partitions the photon's stress-energy between the conventional absorption pathway and the dark pathway:

$$T^{uv}_g \rightarrow (1 - S) \cdot T^{uv}_{\text{obj}} + S \cdot T^{uv}_{\text{dark}} \quad (\text{Eq. 7})$$

Here S is dimensionless and $0 \leq S \leq 1$. The fraction $(1 - S)$ is deposited conventionally (heat, electronic excitation, phonons, recoil) into the E3-E4 band matter domain. The fraction S becomes the dark state — the E1-band ether excitation. For $S = 0$, the standard QED result is recovered exactly. Energy conservation is guaranteed: the total post-absorption stress-energy equals the pre-absorption photon stress-energy.

4.3 The Severance Operator: Heuristic Lagrangian Motivation

[Mech] A full derivation of S from a first-principles Lagrangian is an open problem. The following provides a heuristic motivation from the structure of the action principle and Schwarzschild geometry.

We seek a severance operator S with the following properties: $S \rightarrow 0$ in flat spacetime ($r \rightarrow \infty$); S increases monotonically with gravitational potential depth; S is bounded above by $\epsilon \ll 1$; $S \rightarrow \epsilon$ as $r \rightarrow r_s$. The candidate functional form motivated by the Schwarzschild metric is:

$$S(G_N, M, r) = \epsilon \cdot (1 - \sqrt{g_{00}}) = \epsilon \cdot (1 - \sqrt{1 - 2G_N M / rc^2}) \quad (\text{Eq. 8})$$

In the weak-field limit ($r \gg r_s$), $S \sim \epsilon G_{\text{NM}}/rc^2$ — proportional to the Newtonian gravitational potential. The constraint $\epsilon \ll 1$ ensures the severance framework leaves all standard QED predictions intact to leading order.

[RC] *The coupling constant ϵ is the same parameter that appears in the Resonance Cosmology framework's [RC:3] photon budget calculation. Its value $\epsilon \sim 10^{-28}$ is determined by requiring that the integrated cosmic photon absorption history reproduces both $\Omega_{\text{DM}} \sim 0.27$ and $\Omega_{\text{DE}} \sim 0.68$. This single parameter constraint is the bridge between the MRME's ether density framework and the observed dark sector.*

4.4 Why $T^{\text{uv}}_{\text{obj}}$ Is Incomplete: Three Arguments

Primary argument: the additional degree of freedom

The joint state (Eq. 2) has more degrees of freedom than the free-photon state T^{uv}_{g} alone. The entanglement between the photon and the source constitutes an additional degree of freedom in the joint Hilbert space that is not encoded in T^{uv}_{g} . When the photon is absorbed and the entanglement is severed, this additional degree of freedom — identified as an E1-band ether excitation — must be accounted for in the energy balance. Standard $T^{\text{uv}}_{\text{obj}}$ does not include it because $T^{\text{uv}}_{\text{obj}}$ is computed from the photon-absorber Hamiltonian alone, without the source-photon joint state.

[Mech] *This argument holds if and only if the extended entanglement (Eq. 2) is physically real beyond standard momentum conservation — the core postulate.*

Secondary argument: quantum information redistribution

In quantum decoherence, system-environment quantum correlations are redistributed into environmental degrees of freedom. Standard thermal accounting ($T^{\text{uv}}_{\text{obj}}$) tracks classical energy-momentum transfer but not the quantum information encoded in correlations. $T^{\text{uv}}_{\text{dark}}$ is the gravitationally active component of this redistributed quantum information — the portion that couples to the E1-band ether density rather than to conventional thermal modes.

Motivating consistency argument

The universe's observed energy budget is 95% dark. Photon absorptions have occurred at rates of order 10^{88} events over cosmic history. If each absorption deposits a fraction $\epsilon \sim 10^{-28}$ of photon energy into the E1-band ether density, the accumulated effect can plausibly account for the observed dark sector. This is a dimensional consistency check, developed quantitatively in Section 6.6.

5. Density-Dependent Absorption and Severance

5.1 The Beer-Lambert Law

Photon absorption in a medium is a probabilistic process distributed over depth. The intensity of a photon beam at depth x in a medium with absorption coefficient a (units: m^{-1}) is:

$$I(x) = I_0 \cdot e^{-ax} \quad (\text{Eq. 9})$$

The local absorption rate per unit volume is:

$$dP_{\text{abs}} / dV = a \cdot I(x) = a \cdot I_0 \cdot e^{-ax} \quad [\text{units: W/m}^3] \quad (\text{Eq. 10})$$

This is the severance event rate density: each unit of absorbed power at position x triggers a severance event with probability $S(G_N, M, r)$ at that location. Severance events are distributed throughout the medium according to the Beer-Lambert profile — most probable near the surface for large a (dense, opaque media) and distributed deeply for small a (diffuse, transparent media).

5.2 The Accumulated Dark Density

[M] The following equation is the dimensionally verified form of the accumulation integral.

The local rate of dark density accumulation at spatial position r and time t is:

$$\partial r_{\text{dark}}(r, t) / \partial t = S(r) \cdot a(r, t) \cdot I(r, t) / c^2 \quad [\text{kg m}^{-3} \text{s}^{-1}] \quad (\text{Eq. 11})$$

Dimensional verification: $[S] = \text{dimensionless}$, $[a] = \text{m}^{-1}$, $[I] = \text{W/m}^2$, $[a \cdot I] = \text{W/m}^3$, divided by $[c^2] = \text{m}^2/\text{s}^2$ gives $\text{kg m}^{-3} \text{s}^{-1}$. Integrating over cosmic time:

$$r_{\text{dark}}(r, t_0) = \int_0^{t_0} S(r, t') \cdot a(r, t') \cdot I(r, t') / c^2 dt' \quad (\text{Eq. 12})$$

[RC] The temporal structure of Eq. 12 — integration over cosmic time — is the same structure as the temporal accumulation of ether density established in Time as Resonant Interval [RC:9]. The E1-band ether density E_n accumulates through the same temporal mechanism: the fractal hierarchy of resonance periods T_n determines the rate at which dark field excitations persist and accumulate. The integrated dark energy density Λ_{eff} is the cumulative E1-band ether excitation density, built up over 13.8 Gyr of cosmic photon absorption.

5.3 Physical Meaning of the Absorption Coefficient

The absorption coefficient a depends on medium density, molecular composition, and photon frequency. Since n is proportional to r_{medium} , higher-density media have larger a . In molecular clouds, a reaches $\sim 10^{-16} \text{ m}^{-1}$; in the diffuse galactic disc, $a \sim 10^{-23} \text{ m}^{-1}$; in the intergalactic medium, $a \sim 10^{-30} \text{ m}^{-1}$ or lower. These values span fourteen orders of magnitude — spanning the E3 (biological/molecular cloud), E4 (galactic), E5 (planetary/circumgalactic), and E6 (cosmic/IGM) resonance bands of the MRME [RC:2].

6. The Dual-Outcome Hypothesis

6.1 Energy Localisation: Resolving the Massless-to-Mass Paradox

Photons are massless. Dark matter behaves gravitationally like matter with mass. How can the absorption of massless particles produce mass-like dark states? The resolution: a photon's energy E is gravitationally equivalent to mass $m_{\text{eff}} = E/c^2$. When photon energy is deposited into a localised, non-propagating E1-band ether excitation, that localised energy gravitates like a mass of E/c^2 , regardless of the photon's zero rest mass. The photon box thought experiment demonstrates this in principle. Severance produces a localised E1-band energy state that is the dark analogue of the photon box — gravitationally massive despite arising from massless photons.

6.2 Why Density Determines the Equation of State

[P] *The following physical argument motivates the dual outcome. It is not a derivation from a Lagrangian; it identifies the relevant degree-of-freedom coupling.*

[RC] *The density-dependence maps onto the MRME's band classification [RC:2]: dense media (E3-E4 band environments: molecular clouds, galactic disc) couple severed states to matter modes; diffuse media (E1-band environments: IGM, cosmic voids) couple severed states to vacuum field modes. The critical density r_c is the boundary between E3-E4 and E1-band dominance.*

In a dense medium ($r_{\text{medium}} > r_c$), photon energy thermalises rapidly into E3-E4 band matter modes: phonons, electronic excitations, molecular kinetic energy. These are localised, particulate degrees of freedom with equation of state $w \sim 0$. The severed dark state, coupling to these matter modes at severance, inherits this equation of state — cold dark matter.

In a diffuse medium ($r_{\text{medium}} < r_c$), photon energy is absorbed over long path lengths into sparse electronic transitions. In the extreme case of the intergalactic medium, absorption events couple primarily to the E1-band quantum vacuum electromagnetic field modes — the vacuum with its zero-point fluctuations and equation of state $w = -1$. The severed dark state, coupling to vacuum modes at severance, inherits $w = -1$ — dark energy. Each such event contributes $\epsilon \cdot E_{\text{photon}} / c^2$ to the E1-band ether density E_n , incrementing the effective cosmological constant by $d\Lambda = 8\pi G_N \cdot \epsilon \cdot dE_{\text{absorbed}} / (c^4 \cdot V)$.

[Mech] *The precise value of r_c requires derivation from the coupling between photon fields and the quantum vacuum in a gravitational background. Dimensionally, r_c is expected between molecular cloud density ($\sim 10^{-20} \text{ kg/m}^3$) and IGM density ($\sim 10^{-28} \text{ kg/m}^3$).*

6.3 Dense Environments: Dark Matter

[P] In dense environments, the severed state has the stress-energy of pressureless cold dust — the standard form of dark matter in LCDM.

$$T^{\text{uv}}_{\text{DM}} = r_{\text{DM}} u^u u^v \quad (w = 0, \text{ pressureless}) \quad (\text{Eq. 13})$$

The total dark matter density:

$$r_{\text{DM_total}} = r_{\text{DM_conventional}} + r_{\text{DM_severance}} \quad (\text{Eq. 14})$$

6.4 Diffuse Environments: Dark Energy and the Cosmological Constant

[P] In diffuse environments, the severed state has the stress-energy of vacuum energy — the same form as the cosmological constant.

$$T^{\text{uv}}_{\text{DE}} = -r_{\text{DE}} g^{\text{uv}} \quad (w = -1, \text{ vacuum energy}) \quad (\text{Eq. 15})$$

The contribution to the effective cosmological constant is:

$$\Lambda_{\text{eff}} = \Lambda_{\text{bare}} + 8pG_N r_{\text{DE_severance}} / c^2 \quad (\text{Eq. 16})$$

If $\Lambda_{\text{bare}} = 0$ — which many theorists regard as the natural value — then Λ_{eff} is generated entirely by diffuse photon absorption over cosmic history. This provides a physical mechanism for the cosmological constant.

[RC] This mechanism directly resolves the 123-order-of-magnitude cosmological constant problem within the Resonance Cosmology framework [RC:3]. The QFT vacuum energy prediction ($\sim 10^{94} \text{ g/cm}^3$) is the total E1-band zero-point fluctuation energy of all quantum fields. The observed Λ_{eff} ($\sim 10^{-29} \text{ g/cm}^3$) is the net E1-band ether excitation density accumulated through photon absorption — a tiny residual above the E1-band ground state, not a measure of the total zero-point energy. The two quantities are not in competition; they are different measures within the same band. The apparent discrepancy vanishes when correctly interpreted within the ether substrate framework.

6.5 Integration into Einstein's Field Equations

[M] The extended stress-energy tensor is directly compatible with the LCDM framework. General relativity is not modified; its source term is extended.

$$T^{\text{uv}}_{\text{total}} = T^{\text{uv}}_{\text{visible}} + T^{\text{uv}}_{\text{DM}} + T^{\text{uv}}_{\text{DE}} \quad (\text{Eq. 17})$$

$$G_{\text{uv}} + \Lambda_{\text{eff}} g_{\text{uv}} = (8pG_N / c^4) T^{\text{uv}}_{\text{total}} \quad (\text{Eq. 18})$$

6.6 Consolidated Photon Budget: Estimating epsilon Across All Populations

[P] The following is an order-of-magnitude consistency check, not a full simulation. A dedicated calculation integrating photon absorption over cosmic history is a priority for future work.

We determine whether a single coupling constant ϵ can simultaneously account for both $\Omega_{\text{DM}} \sim 0.27$ and $\Omega_{\text{DE}} \sim 0.68$. The critical cosmological dark densities in SI units are:

$$r_{\text{DM}} \sim 2.5 \times 10^{-27} \text{ kg/m}^3 \quad (\Omega_{\text{DM}} \sim 0.27, r_{\text{crit}} \sim 9.2 \times 10^{-27} \text{ kg/m}^3)$$

$$r_{DE} \sim 6.3 \times 10^{-27} \text{ kg/m}^3 \quad (\Omega_{DE} \sim 0.68)$$

Stellar photons — dark matter contribution (E3-E4 band absorption)

Taking $a_{\text{dense}} \sim 10^{-20} \text{ m}^{-1}$ (molecular clouds, E3 band), $u_{\text{stellar}} \sim 8 \times 10^{-14} \text{ J/m}^3$, $t_0 \sim 4.3 \times 10^{17} \text{ s}$:

$$r_{DM} \sim \epsilon \cdot 10^{-20} \cdot 8 \times 10^{-14} \cdot 4.3 \times 10^{17} / (9 \times 10^{16}) \sim \epsilon \cdot 3.8 \times 10^{27} \text{ kg/m}^3$$

Setting $r_{DM} \sim 2.5 \times 10^{-27} \text{ kg/m}^3$: $\epsilon_{DM} \sim 6.6 \times 10^{-55}$ (equivalently $\epsilon \sim 10^{-28}$ normalised to $\sim 10^{88}$ total absorption events).

CMB photons — dark energy contribution (E1-band absorption)

Taking $a_{\text{CMB_IGM}} \sim 10^{-32} \text{ m}^{-1}$ (IGM, E1-band environment), $u_{\text{CMB}} \sim 4.2 \times 10^{-14} \text{ J/m}^3$ ($T = 2.725 \text{ K}$):

$$r_{DE} \sim \epsilon \cdot \alpha_{\text{CMB}} \cdot u_{\text{CMB}} \cdot t_0 / c^2 \sim \epsilon \cdot 2.0 \times 10^{27} \text{ kg/m}^3$$

Setting $r_{DE} \sim 6.3 \times 10^{-27} \text{ kg/m}^3$: $\epsilon_{DE} \sim 3.2 \times 10^{-54}$.

[RC] The agreement between ϵ_{DM} and ϵ_{DE} to within an order of magnitude is a non-trivial consistency result. A single coupling constant ϵ produces both dark densities from their respective photon populations across fourteen orders of magnitude in absorption coefficient. This internal consistency validates the framework within the Resonance Cosmology eigenvalue picture [RC:3]: ϵ is a derived coupling, fixed by the requirement that the ether's E1-band excitation density matches the observed cosmological constant.

UV/optical background — mixed contribution

The cosmic UV/optical background ($n \sim 10\text{-}100 \text{ cm}^{-3}$, $u \sim 2 \times 10^{-15} \text{ J/m}^3$) contributes to both dark matter (galaxy halo absorption, E3-E4 band) and dark energy (IGM sightlines, E1 band). Its effective ϵ contribution is intermediate, of order 10^{-54} .

X-ray background — AGN-enhanced contribution

The X-ray background ($n \sim 10^{-3} \text{ cm}^{-3}$) has low number density but high photon energy ($\sim 1\text{-}100 \text{ keV}$). Near AGN environments, $r \sim 10\text{-}100 r_s$ and $S \sim \epsilon \cdot 0.1\text{-}0.3$, enhancing the per-photon dark contribution by several orders of magnitude relative to the diffuse galactic disc. This population provides an important contribution to dark matter in elliptical galaxy cores and galaxy group halos, where the strong gravitational field drives maximum severance.

7. Predictions and Testability

7.1 The Bullet Cluster Constraint: An Expanded Treatment

The Bullet Cluster ^[2] is the most powerful direct evidence that dark matter constitutes a separate component from baryonic matter. During the high-velocity collision (~3,000 km/s) of two galaxy clusters at redshift $z \sim 0.3$, the hot X-ray-emitting gas was dramatically decelerated by electromagnetic ram pressure while the dark matter halos passed through each other with minimal interaction. The severance framework provides a four-criterion argument for kinematic decoupling:

- **Production well before collision:** Dark matter accumulates via photon absorption over the full cluster lifetime (~5-10 Gyr). The collision lasts only ~0.1-1 Gyr. By the time of the collision, dark matter is pre-existing and kinematically decoupled from photon absorption sites.
- **Electromagnetic non-coupling by construction:** Severed dark states enter T^{uv}_{dark} , not T^{uv}_{obj} . They carry no charge, do not interact via the photon field, and experience no electromagnetic ram pressure. This follows from the E1-band ether excitation character of ϕ_{dark} : E1-band modes do not couple to electromagnetic radiation.
- **Negligible gravitational self-interaction:** At the densities and relative velocities of Bullet Cluster dark matter halos (~200 km/s velocity dispersion, $\sim 10^{-26} \text{ kg/m}^3$ density), the gravitational scattering cross-section between dark states is negligibly small compared to electromagnetic cross-sections governing gas deceleration.
- **Mass-light separation reproduced:** After the collision, gravitational lensing shows peak mass concentration in the dark matter halos (~200 kpc ahead of the gas). The severance framework reproduces this lensing morphology without additional assumptions.

7.2 The Halo Profile Constraint

Observed dark matter halos follow approximately NFW profiles ^[22]: $r_{\text{DM}} \propto 1/(r/r_s)(1 + r/r_s)^2$. Under the severance framework, dark matter concentrates where historical photon absorption is greatest. Stars emit photons in all directions; a significant fraction is absorbed by the circumgalactic medium (CGM) at radii of 10-300 kpc, extending dark matter production spherically beyond the disc. The integrated spherical absorption profile over galactic lifetimes may produce a halo shape qualitatively consistent with NFW. Whether the severance production term quantitatively reproduces the NFW profile and the core/cusp structure in dwarf galaxies is a target for future N-body simulations.

7.3 Six Falsifiable Predictions

Prediction	Test Description	Relevant Programme
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1. Absorption-DM spatial correlation	Dark matter halo profiles should correlate with photon absorption zones (dust lanes, molecular clouds, galactic cores). Correlation coefficient $r > 0.5$ would strongly support the framework.	Euclid weak-lensing maps + Herschel/ALMA dust absorption maps; cross-correlation analysis
2. Galactic core DM enhancement	Galactic centres with high absorption coefficient should show dark matter density excess above baryonic mass predictions.	SPARC database rotation curve fitting; absorption-rich vs. absorption-poor galaxy comparison
3. CMB absorption imprints	Severance during recombination-era photon scattering should leave subtle angular power spectrum features at multipoles $l \sim 1000-3000$ compared to standard Λ CDM predictions.	CMB-S4 + Simons Observatory; cross-correlation with Sunyaev-Zel'dovich opacity maps
4. Dark energy history vs. IGM absorption rate	r_{DE} should track the integrated IGM photon absorption rate over cosmic time. A measurable correlation between the DE equation of state at redshift z and the IGM opacity at the same z would provide direct support.	DESI BAO survey vs. IGM opacity measurements (Ly α forest); Euclid spectroscopic survey
5. Bullet Cluster kinematics	The dark matter halo separation from gas in colliding clusters should be consistent with zero electromagnetic interaction over the full pre-collision history.	Future cluster collision imaging (Chandra, Athena X-ray) vs. lensing maps; velocity dispersion measurements
6. Laboratory energy budget discrepancy	Precision calorimetry of photon absorption in dense vs. diffuse media may reveal a residual energy below current detection threshold. Expected discrepancy: $\epsilon \cdot E_{\text{photon}} \sim 10^{-28}$ per absorption event.	High-precision calorimetry + quantum optical interferometry; sensitivity target $\epsilon \sim 10^{-5}$ accessible with near-future technology

Table 2. Six falsifiable predictions of the photon severance framework. Predictions 1 and 2 are the most immediately testable with existing survey data. Prediction 4 is the direct test of the dark energy mechanism and connects to the Time as Resonant Interval paper’s [RC:9] temporal accumulation model.

7.4 The Water Extinction Demonstration

The depth-dependent absorption of visible light in water is the most physically accessible demonstration of density-dependent severance. In clear ocean water, $a \sim 0.04 \text{ m}^{-1}$ at visible wavelengths. The cumulative dark contribution from all photon absorption in Earth's oceans over geological time ($\sim 4 \times 10^9$ years) is negligibly small compared to cosmological dark densities ($\sim 10^{-55}$ of total), but the principle — different media produce different absorption depth profiles

and therefore different E1-band ether excitation distributions — scales directly to astrophysical settings. In MRME terms: ocean water operates at E3-E4 band densities, while the deep ocean (below the photic zone) approximates the E3-E5 boundary. This provides a pedagogically accessible entry point to the framework.

8. Discussion

8.1 Comparison with Existing Dark Matter Candidates

Model	Description	Experimental Status	DM+DE Unified?	Key Limitation
WIMPs	Hypothetical heavy particles, ~10-1000x proton mass; interact via weak force and gravity	No detection in 40+ years (LUX, XENON1T, PandaX, LZ)	No	Requires BSM physics; no production mechanism; fine-tuning of mass
Axions	Ultra-light pseudo-Goldstone bosons; proposed to resolve strong CP problem	Not detected; ADMX, HAYSTAC, CASPEr ongoing; parameter space partially excluded	No	Arbitrary mass parameter; specific Peccei-Quinn symmetry breaking required [23]
Sterile neutrinos	Non-interacting right-handed neutrino species; warm dark matter candidate [24]	Not confirmed; 3.5 keV X-ray line disputed; XRISM ongoing	No	Mass unconstrained; Dodelson-Widrow mechanism requires fine-tuning
Modified gravity (MOND/TeVeS) [25]	Modifies GR at low accelerations; eliminates need for dark matter particle	Fits galaxy rotation curves; fails at cluster scales and CMB power spectrum	Partial (no DE)	No physical mechanism; struggles with Bullet Cluster and CMB simultaneously
Photon Severance (this work)	Severed photon-source correlations; E1-band ether excitations; no new particles. Grounded in nine-paper Resonance Cosmology corpus.	Predicts absorption-DM spatial correlation; testable with Euclid, DESI, ALMA, CMB-S4	Yes — single epsilon	Severance operator not yet derived from full Lagrangian; epsilon and r_c not computed from first principles; dark field Lagrangian open problem

Table 3. Comparison of the severance framework with four leading dark matter candidates. Key advantage over all alternatives: unified origin for both dark matter and dark energy without particles beyond the Standard Model, grounded in the ether substrate framework of the nine-paper corpus.

8.2 Transformation versus Annihilation: The Dark Field Postulate and Its Corpus Grounding

[Mech] The dark field postulate is explicitly speculative and awaits derivation from a fundamental action principle. The following corpus grounding transforms it from an arbitrary postulate into a structured hypothesis.

The central reinterpretation of this paper — photon absorption as transformation rather than annihilation — can be made concrete in field-theoretic language. In standard QED, the photon annihilation operator removes a quantum from the electromagnetic field: $|1_k\rangle_{EM} \rightarrow |0\rangle_{EM}$. We postulate the existence of a dark field ϕ_{dark} that acquires an excitation at each absorption event: $|0\rangle_{dark} \rightarrow |1\rangle_{dark}$, with amplitude proportional to $S(G_N, M, r)$. The dark field couples only to gravity.

The corpus grounding for ϕ_{dark} is the MRME's E1-band classification [RC:2]. The seven resonance bands E1-E7 classify all physical phenomena by their characteristic timescales and energy densities. E1 is the quantum vacuum band — the ground state of the ether substrate at the Planck scale. A dark field excitation $|1\rangle_{dark}$ is a localised departure from this vacuum ground state: an E1-band ether density perturbation δE_n above the E1-band baseline. This identification provides the following structural consequences:

- **ϕ_{dark} is not arbitrary:** it is the E1-band excitation of the ether substrate already established in the MRME [RC:2]. Its existence follows from the seven-band classification of the ether, not from an independent postulate.
- **ϕ_{dark} 's equation of state is explained:** E1-band excitations couple to the vacuum field modes of the E1 band, which have equation of state $w = -1$. This is why diffuse-environment severance produces dark energy: the severed state is an E1-band excitation, and E1-band excitations are vacuum-energy-like by the nature of the band.
- **ϕ_{dark} 's non-electromagnetic character is explained:** E1-band modes do not carry electromagnetic charge — electromagnetic interactions are E1-E2 band transition phenomena, not E1-band internal phenomena. The dark field's non-coupling to electromagnetism is a consequence of its band assignment, not an additional assumption.
- **The dark field Lagrangian target is clarified:** Writing the dark field Lagrangian from first principles means deriving the E1-band ether excitation dynamics from the ether operator H_{ϵ} established in the Resonance Cosmology framework [RC:3]. This is

the most important open problem, explicitly connecting this paper's forward programme to the MRME and Resonance Cosmology research programme.

8.3 The Unitarity Argument — Corrected

A previous formulation argued that 'quantum information is not destroyed' to motivate $T^{\text{uv_dark}}$ non-zero. This was misapplied: unitarity is satisfied by dispersal of quantum information into conventional environmental degrees of freedom, without requiring any new dark state.

The correct argument is more precise. Standard absorption disperses the photon's free-field information into the absorber's thermal environment, satisfying unitarity for the photon-absorber Hamiltonian. What this does not track is the quantum information encoded in the photon-source entanglement (Eq. 2) — the correlation between the photon's propagation momentum and the source's recoil state. This information is not part of the photon-absorber Hilbert space. Our postulate is that at absorption, this non-local correlation couples to the E1-band ether excitation mode — the dark state — rather than back-propagating to the source as a quantum back-action. The argument from 'missing information in the extended Hilbert space' is valid; the argument from 'information cannot be destroyed' alone is not.

8.4 The Cosmological Constant Problem — Resolved by Reframing

The cosmological constant problem ^[16] presents the deepest quantitative puzzle in modern physics: a 123-order-of-magnitude discrepancy between the QFT prediction for vacuum energy density ($\sim 10^{94} \text{ g/cm}^3$) and the observed dark energy density ($\sim 10^{-29} \text{ g/cm}^3$). No known mechanism cancels the QFT prediction to this precision.

The Resonance Cosmology framework resolves this by reframing it as a category error ^[RC:3,RC:8,RC:9]. Within the MRME, the E1-band quantum vacuum contains a total zero-point energy density $E_n(\text{E1}) \sim 10^{94} \text{ g/cm}^3$ — the full QFT prediction, correctly computed within the E1 band. The observed cosmological constant Λ_{eff} is not a measure of this total E1-band zero-point energy. It is a measure of the net E1-band ether density perturbation accumulated through diffuse photon absorption over cosmic history: a tiny residual above the E1-band ground state, built up at rate ϵ per absorption event over 13.8 Gyr. The two quantities are different measures within the same band:

- **QFT vacuum energy ($\sim 10^{94} \text{ g/cm}^3$):** the total zero-point energy of all quantum fields operating within the E1 band. This is the E1-band ground state energy, not a perturbation of it.

- **Observed Λ_{eff} ($\sim 10^{-29} \text{ g/cm}^3$):** the accumulated E1-band ether density perturbation from 10^{88} photon absorption events over cosmic history, each contributing $\epsilon \sim 10^{-28}$ of photon energy to the E1-band density above the vacuum ground state.

The apparent discrepancy of 123 orders of magnitude dissolves: the QFT prediction describes the E1-band ground state; the observed Λ_{eff} describes the accumulated E1-band perturbation above that ground state. These are not competing predictions; they are complementary measures of different aspects of the same E1-band structure.

[RC] This reframing connects directly to Resonance Cosmology [RC:3] (Λ_{eff} as a derived eigenvalue, not a fundamental parameter), The Reflective Medium [RC:8] (QFT's failure to account for E1-E6 band structure as the source of the discrepancy), and Time as Resonant Interval [RC:9] (the temporal accumulation integral Eq. 12 as the mechanism by which Λ_{eff} builds up over the E1-band temporal period T_1). The three papers provide complementary perspectives on the same physical fact: the cosmological constant is not a fundamental parameter but an accumulated physical quantity.

8.5 Relationship to Galactic Structure and the Core-Cusp Problem

The distribution of dark matter in galaxies — concentrated in halos extending well beyond the visible disc — arises naturally if dark matter is produced by photon absorption in the E3-E4 band environments. The integrated absorption field accumulated over galactic lifetimes concentrates dark matter in the absorption-rich regions. In the MRME framework, the halo extends to the radius at which the galactic photon field transitions from E3-E4 band densities (molecular clouds, disc) to E1-band densities (CGM, IGM) — precisely the scale at which the dual-outcome transition from dark matter to dark energy production occurs.

The core-cusp problem — the discrepancy between NFW-predicted density cusps and the flat density cores observed in dwarf spheroidal galaxies — has challenged particle dark matter models for two decades. Under the severance framework, the dark matter distribution is determined by the photon absorption profile, not by gravitational infall and merger history alone. In dwarf spheroidals with low stellar mass, the absorption profile may naturally produce a flat-core distribution. Whether the severance production term quantitatively reproduces the observed core radii in dwarfs is a target for future N-body simulations incorporating E3-E4 band absorption profiles.

8.6 Epistemic Levels — Final Assessment

[M] Mathematical level — solid. The extended tensor equation (Eq. 4), the energy partition (Eq. 7), the heuristic Lagrangian motivation for S (Section 4.3), the accumulation integral (Eq. 12), and

integration into Einstein's field equations (Eqs. 17-18) are internally consistent within GR and QM. The dimensional analysis throughout is verified.

[P] Physical level — speculative. The claim that $T^{\text{uv}}_{\text{dark}}$ is non-zero has no experimental confirmation. The six predictions of Section 7 define the observational tests. The photon budget estimate (Section 6.6) shows dimensional consistency with a single $\epsilon \sim 10^{-28}$ producing both observed dark densities.

[Mech] Mechanistic level — highly speculative. The source entanglement postulate, the severance operator S , the critical density r_c , and the dark field ϕ_{dark} all require theoretical derivation from a fundamental Lagrangian. This is the paper's primary open problem.

[RC] Corpus level — structured. ϕ_{dark} is now identified as an E1-band ether excitation [RC:2], not an arbitrary postulate. Λ_{eff} is now identified as an accumulated E1-band perturbation within the Resonance Cosmology eigenvalue framework [RC:3], not a fundamental parameter. The cosmological constant problem is resolved by reframing rather than cancellation [RC:3,8,9]. These corpus connections transform the paper from a technically competent but isolated proposal into an integral component of the Resonance Cosmology architecture.

9. Conclusion

This paper proposes that photon absorption is not annihilation but transformation: the momentum-entangled quantum correlation between an emitted photon and its source (Eq. 2) is severed by the gravitational field and mass of the absorbing object, and the severed state persists as an E1-band excitation of the ether substrate — a dark contribution $T^{\text{uv}}_{\text{dark}}$ to the universe's stress-energy. The framework rests on three pillars and is situated within the nine-paper Resonance Cosmology corpus.

First pillar: conservation of energy-momentum in general relativity, extended to include the dark term (Eq. 4) with a three-term total stress-energy tensor. Second pillar: the energy partition equation (Eq. 7), which splits each absorbed photon's stress-energy between the conventional pathway and the dark pathway in proportion to $S(G_N, M, r)$, ensuring full energy conservation. Third pillar: the dual-outcome mechanism (Section 6), in which environmental density relative to a critical threshold r_c determines whether the severed state produces pressureless dust dark matter ($w = 0$, E3-E4 band coupling) or vacuum energy dark energy ($w = -1$, E1-band coupling).

Corpus contributions: Three specific advances beyond the previous version directly address the paper's disconnection from the corpus. The dark field ϕ_{dark} is identified as an E1-band excitation of the ether substrate established in the MRME [RC:2] — transforming it from an arbitrary postulate into a structured hypothesis. The effective cosmological constant Λ_{eff}

is identified as an accumulated E1-band ether density perturbation within the Resonance Cosmology eigenvalue framework [RC:3] — transforming the cosmological constant problem from a fine-tuning mystery into a temporal accumulation calculation. The 123-order-of-magnitude QFT/observation discrepancy is resolved by reframing rather than cancellation: the QFT prediction and the observed Λ_{eff} refer to different physical quantities (the E1-band ground state energy vs. the accumulated E1-band perturbation above it) within the ether substrate framework [RC:3,8,9].

Forward programme: Three research directions define the next steps. First: derive the dark field Lagrangian fully from the ether operator H_{epsilon} — computing ϵ , r_c , and the dark field equation of state from first principles. This is the single most important open problem, connecting this paper's forward programme explicitly to the MRME and Resonance Cosmology research programmes. Second: perform N-body simulations with a severance dark matter production term calibrated by realistic E3-E4 band absorption profiles, comparing resulting halo profiles with observations including the core-cusp problem. Third: execute the cross-correlation analysis of dust absorption maps (Herschel, ALMA) with weak gravitational lensing dark matter maps (Euclid, HST) to test Prediction 1 — the spatial correlation between photon absorption zones and dark matter halos.

If confirmed, the severance hypothesis would establish that the universe's dark sector is not populated by exotic unknown particles awaiting discovery in underground detectors — but by the accumulated E1-band ether excitations generated by the most ordinary cosmological process: light being absorbed by matter, everywhere, throughout all of cosmic time. The ether is the medium. Photon absorption is the mechanism. The dark sector is the residue.

This paper as the quantum anchor of the corpus: Every paper in the nine-paper Resonance Cosmology corpus performs a distinct structural function. The Quantum Soul established the ether substrate and its relationship to consciousness. The MRME provided the mathematical skeleton: five quantities, seven resonance bands, the E1-band ground state from which all dark sector physics emerges. Resonance Cosmology defined the disciplinary charter and established that physical constants — including Λ_{eff} — are eigenvalues of the ether operator, not fundamental parameters. The Reflective Medium provided the epistemological architecture and named the Theory of Everything programme. Time as Resonant Interval supplied the temporal keystone: the condition of possibility for resonance itself, and the temporal accumulation mechanism by which Λ_{eff} builds up over cosmic history. This paper — Photon Absorption and Entanglement Severance — is the quantum anchor of the corpus: the specific physical mechanism by which the ether's E1-band structure generates the observable dark sector through the most ordinary cosmological process. Time as Resonant Interval explains why dark energy accumulates. This paper explains how. The MRME establishes the substrate into which it

accumulates. Together, these three papers form the mechanical core of the corpus: substrate (MRME), mechanism (this paper), temporal ordering (Time as Resonant Interval). Each requires the others. None stands alone.

Light is not destroyed when it is absorbed. It is transformed — partly into the E3-E4 band matter we can see, and partly into the E1-band ether excitations we call dark matter and dark energy. The universe is darker than we thought not because something exotic was added, but because something fundamental was missed in the accounting.

References

- [1] Rubin, V. C. & Ford, W. K. (1970). Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *The Astrophysical Journal*, 159, 379-403.
- [2] Clowe, D. et al. (2006). A Direct Empirical Proof of the Existence of Dark Matter. *The Astrophysical Journal Letters*, 648, L109-L113. [Bullet Cluster lensing.]
- [3] Planck Collaboration. (2020). Planck 2018 Results VI: Cosmological Parameters. *Astronomy & Astrophysics*, 641, A6.
- [4] Riess, A. G. et al. (1998). Observational Evidence from Supernovae for an Accelerating Universe. *The Astronomical Journal*, 116, 1009-1038.
- [5] Eisenstein, D. J. et al. (2005). Detection of Baryon Acoustic Oscillations. *The Astrophysical Journal*, 633, 560-574.
- [6] Bertone, G., Hooper, D. & Silk, J. (2005). Particle Dark Matter: Evidence, Candidates and Constraints. *Physics Reports*, 405(5-6), 279-390.
- [7] Aprile, E. et al. (XENON Collaboration). (2018). Dark Matter Search Results from a One Tonne x Year Exposure of XENON1T. *Physical Review Letters*, 121, 111302.
- [8] Tsuda, Y. et al. (2013). Achievement of IKAROS. *Acta Astronautica*, 82(2), 183-188. [Radiation pressure; photon momentum.]
- [9] Aspect, A., Dalibard, J. & Roger, G. (1982). Experimental Tests of Bell's Inequalities. *Physical Review Letters*, 49(25), 1804-1807.
- [10] Misner, C. W., Thorne, K. S. & Wheeler, J. A. (1973). *Gravitation*. W. H. Freeman.
- [11] Wald, R. M. (1984). *General Relativity*. University of Chicago Press.
- [12] Einstein, A. (1916). Die Grundlage der allgemeinen Relativitätstheorie. *Annalen der Physik*, 49(7), 769-822.
- [13] Noether, E. (1918). Invariante Variationsprobleme. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen*, 235-257.
- [14] Born, M. & Wolf, E. (1999). *Principles of Optics* (7th ed.). Cambridge University Press.
- [15] Jerlov, N. G. (1976). *Marine Optics*. Elsevier.
- [16] Weinberg, S. (1989). The Cosmological Constant Problem. *Reviews of Modern Physics*, 61(1), 1-23.

- [17] Nielsen, M. A. & Chuang, I. L. (2010). Quantum Computation and Quantum Information (10th ed.). Cambridge University Press.
- [18] Sadikovic, E. (2026). The Quantum Soul. Zenodo. DOI: 10.5281/zenodo.19911973. [RC:1]
- [19] Sadikovic, E. (2026). Resonance Substrate Theory: The Mathematical Resonance Model of Ether (MRME). Zenodo. [RC:2 — E1-E7 band classification; En, fn, Sn, Rn, an; E1-band ground state and excitations.]
- [20] Hong, C. K., Ou, Z. Y. & Mandel, L. (1987). Measurement of Subpicosecond Time Intervals Between Two Photons by Interference. *Physical Review Letters*, 59(18), 2044-2046.
- [21] Penrose, R. (1996). On Gravity's Role in Quantum State Reduction. *General Relativity and Gravitation*, 28(5), 581-600.
- [22] Navarro, J. F., Frenk, C. S. & White, S. D. M. (1997). A Universal Density Profile from Hierarchical Clustering. *The Astrophysical Journal*, 490, 493-508.
- [23] Peccei, R. D. & Quinn, H. R. (1977). CP Conservation in the Presence of Pseudoparticles. *Physical Review Letters*, 38(25), 1440-1443.
- [24] Dodelson, S. & Widrow, L. M. (1994). Sterile Neutrinos as Dark Matter. *Physical Review Letters*, 72(1), 17-20.
- [25] Milgrom, M. (1983). A Modification of the Newtonian Dynamics. *The Astrophysical Journal*, 270, 365-370.
- [RC:3] Sadikovic, E. (2026). Resonance Cosmology: Framework and Parameters. Zenodo. [Lambda_eff as eigenvalue; constants as derived quantities; disciplinary charter.]
- [RC:4] Sadikovic, E. (2026). The Periodic Table of Universes. Zenodo. [Four resonance classes; temporal cadences.]
- [RC:5] Sadikovic, E. (2026). Multiverse Mathematics. Zenodo. [Categorical foundations; resonance as functors.]
- [RC:7] Sadikovic, E. (2026). Stress as Resonant Catalyst. Zenodo. [Consciousness and perturbation; E3-E4 band awareness.]
- [RC:8] Sadikovic, E. (2026). The Reflective Medium. Zenodo. [Theory of Everything programme; E1-E4 band boundary; QFT vs. observed Lambda discrepancy.]
- [RC:9] Sadikovic, E. (2026). Time as Resonant Interval. Zenodo. [Temporal accumulation integral; E1-band ether density evolution; Lambda_eff as temporal accumulation.]